Devanshu Kumar Ranjan

+91-6206938457 | devanshukranjan@gmail.com | Patna, India | linkedin | github | portfolio

Summary

AI Engineer at KAS Global Commerce (My Seller Central) and M.Tech Cyber Security graduate from NIT Patna (CGPA: 9.06/10), specializing in production AI systems, LLM agent workflows, and privacy-preserving machine learning. Architect of the enterprise Root Cause Analysis (RCA) Diagnostic Engine: an AI investigator explaining complex advertising performance anomalies in natural language. Engineered scalable multimodal insight pipelines (Gemini 2.5 Flash), high-throughput distributed scrapers (Zyte API @ 1,000 req/min), and vector retrieval services (FastAPI, OpenAI embeddings, PostgreSQL + `pgvector`). Master's thesis on locally differentially private (LDP) federated sequential recommendation. Experienced in designing resilient, low-latency AI backends with Python, FastAPI, PyTorch, LangGraph, and PostgreSQL.

Education

National Institute of Technology, Patna — Patna, India
M.Tech., Cyber Security (CSE) — CGPA: 9.06/10 — *August 2024 – May 2026*

Graphic Era University, Dehradun — Dehradun, India
B.Tech., Computer Science and Engineering — CGPA: 8.83/10 — *June 2019 – June 2023*

Sri Chaitanya Techno School (CBSE) — Visakhapatnam, India
Senior Secondary — 80.6% — *2019*

Cambridge School (ICSE) — Cuttack, Odisha
Secondary — 90.17% — *2017*

Experience

KAS Global Commerce Inc. (My Seller Central) — Bilaspur, HP (On-site)
AI Engineer — *July 2026 – Present*

- **Root Cause Analysis (RCA) Engine (Research to Production):** Architected an end-to-end AI diagnostics system analyzing Amazon advertising performance. Integrated as the `rca_analytics` agent under a LangGraph multi-agent orchestrator, translating natural language queries (English and Hinglish) into automated anomaly investigations; shipped **7 live diagnostics** across **50+ root-cause rules** (bid shifts, Buy Box loss, budget exhaustion) with dynamic per-account statistical thresholding ($\mu + 1.5\sigma$), dual baseline windows, and automated negative keyword recommendations with over-blocking protection.
- **Automated Monthly Pipeline Ingestion & Multimodal Insights:** Automated the production catalog intelligence cycle: once the two-stage scraper completes its monthly crawl for a marketplace, multimodal text and image insights (quantitative metrics + Gemini 2.5 Flash Vision) execute automatically via scheduled cron workflows on a monthly basis, requiring code changes only when onboarding new marketplace adapters.
- **Cross-Marketplace Expansion:** Extended distributed scraping and multimodal text/image insight pipelines from initial Amazon and ONDC implementations to Walmart, Flipkart, eBay, and OnBuy; extracted visual archetypes across Amazon's 19 Standard and 21 Premium A+ module templates.
- **Mentorship & Governance:** Actively guiding and mentoring junior engineering teammates on marketplace adapter development, error telemetry monitoring, Zyte rate-limit balancing, and scheduler pipeline health.

KAS Global Commerce Inc. (My Seller Central) — Bilaspur, HP (On-site)
AI Research Intern — *September 2025 – June 2026*

- Built an e-commerce intelligence platform powering automated listing creation, content grading, and chatbot query answering.
- Engineered a two-stage distributed web scraping pipeline across Amazon.in, Amazon.com, and ONDC; reduced full-catalog scraping time from **2–3 months to 1.5 weeks** via threaded Zyte API calls at **1,000 req/min** with MongoDB persistence.
- Developed taxonomy ETL and semantic vector search using OpenAI `text-embedding-3-large` (3,072 dimensions) and PostgreSQL with `pgvector` (IVFFLAT indexing) for cross-marketplace category mapping, achieving sub-50ms retrieval latency.
- Designed a hybrid multimodal insight pipeline combining quantitative statistical analysis with Gemini 2.5 Flash inference per category, using parallel project balancing to optimize rate limits.

Projects

Privacy-Preserving Federated Sequential Recommendation (M.Tech Dissertation) August 2025 – May 2026

- **Tools:** Python, PyTorch, Federated Learning, Differential Privacy, Contrastive Learning
- Researched and built a privacy-preserving sequential recommendation architecture preventing user trajectory leakage.
- Formulated **LDP-SeqMask** (Locally Differentially Private Sequence Masking), injecting calibrated noise into interaction sequences locally so raw client item sequences are never transmitted to central servers.
- Replaced unstable Exponential Mechanism item selection with a **Laplace perturbation mechanism**, eliminating gradient divergence and stabilizing federated model aggregation.
- Integrated **View-based Contrastive Learning (View-CL)** to maintain high recommendation accuracy (NDCG@10) even under heavy differential privacy noise, alongside sequence-aware heterogeneous client model selection.

Multi-Marketplace Vector Search & E-Commerce Scraping API January 2026 – Present

- **Tools:** FastAPI, PostgreSQL, pgvector, OpenAI API, Gemini SDK, Pydantic, Docker
- Developed and containerized production REST APIs for semantic category mapping, multimodal product retrieval, and category-level insight generation serving conversational AI agents.
- Engineered hybrid text and vision category retrieval using Gemini vision labeling and `pgvector` cosine similarity over 100K+ taxonomy breadcrumbs with IVFFLAT index tuning.
- Built robust automated health checks, error handlers, and Swagger/ReDoc interactive documentation.

Autonomous AI-Based Threat Detection and Elimination Engine May 2025 – June 2025

- **Tools:** Python, PyTorch, Scikit-learn, CNN, Tkinter, Pandas
- Built an autonomous AI threat detection engine to identify ransomware signatures and zero-day intrusion patterns.
- Trained and benchmarked CNN and ensemble models on the **UGRansome 2024** and **UNSW-NB15** datasets.
- Developed an interactive GUI-driven network monitoring simulator for real-time packet inspection and automated mitigation.
- GitHub: github.com/devanshukranjan/Autonomous-AI-Based-Threat-Detection-and-Threat-Elimination-Engine.git

Intrusion Detection System via Genetic Algorithm Feature Selection March 2025 – April 2025

- **Tools:** Python, Scikit-learn, XGBoost, NumPy, Pandas
- Engineered an intrusion detection model on the **Bot-IoT** dataset utilizing Genetic Algorithm (GA) feature selection.
- Eliminated redundant network features by over 70% while preserving $>$**99% attack detection recall** using XGBoost.
- GitHub: github.com/devanshukranjan/IDS-Model-using-Genetic-Algorithm-Based-Feature-Selection.git

Technical Skills

Generative AI & LLMs: LangGraph, LangChain, RAG Systems, Gemini 2.5 Flash, OpenAI Embeddings, Prompt Engineering, Multimodal AI, Agentic Workflows

Machine Learning & Deep Learning: PyTorch, Scikit-learn, XGBoost, CNNs, Contrastive Learning, Federated Learning, Locally Differential Privacy (LDP)

Backend & Data Systems: Python, FastAPI, Pydantic, PostgreSQL, `pgvector`, MongoDB Atlas, SQL, REST APIs

Data Engineering & Scraping: Zyte API, BeautifulSoup4, Threaded Ingestion, ETL Pipelines, Pandas, NumPy

Languages & Tools: Python, C++, SQL, Git, GitHub, Docker, Linux/Bash, VS Code, Postman

Certifications

Google Cybersecurity Specialization (Coursera) December 2024 – March 2025

Certified Python Developer Associate (PDAC) March 2024

Data Science for Engineers (NPTEL) January 2022 – March 2022

Machine Learning (Teachnook) February 2022 – March 2022